Collections Copilot

Adda Wale | Finals build | Snapshot 28 September 2026 | **AI** = model output, **RULES** = reviewed code and SQL, **HUMAN** = person decides

Source release: CRM, cards, lending, core banking, collections and bureau data. CSV / Parquet, bilingual notes/transcripts, current policies and optional synthetic WAV audio.

1 Data Product Factory

RULES: DuckDB raw → curated → C360.
Typed checks keep raw values; identity rules hold uncertain links for review.
Outputs: C360, lineage, counted quality report, YAML contract, holiday calendar.

2 Insight / NLP

RULES: reviewed metric compiler → bound SQL (never model-written SQL). Trends, rankings, counts, programmes; one-line summary computed from rows; unsupported → supported rephrasings.
AI (optional): Gemini picks a reviewed plan; local Whisper for audio.

3 Feature Store

RULES: one versioned offline/live definition: case, promise, contact features and nine bilingual hardship cues (POL-COLL-004 §4.2).
Parity checked offline vs live.
Outputs: stored features and registry.

4 Decisioning

RULES: ordered action plan (restriction → representative → hardship programmes → promise → legal contact window → payment ask), work queue tiers, portfolio checks.
AI: logistic classifier flags possible hardship only.
HUMAN: supervisor approves / edits / rejects; audit.

Application: Next.js → FastAPI. Pages: Overview with portfolio checks, Work queue, Customers, Ask, Agent brief, Supervisor review, Data and features. DuckDB analytics; SQLite sessions, reviews and append-only audit. No key means fully local.

Governance across layers: server-side roles and assigned-case access; every step cites policy and source record; protected fields and vulnerability never enter the model or queue order; no automatic customer contact.